

Problem Statement #17 — Vaccination Cohort & Dose Scheduling System

Name: Zayaan Ahmed

SRN: PES1UG24AM457

Deliverable 1: Complete Requirements Table

Functional Requirements

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall allow a Citizen Registrant to book a vaccination slot at a selected center for a specific dose number, date, and time window.	High	Pass: Slot is reserved and a confirmation reference is generated when a valid, available slot is selected. Fail: Booking succeeds for an already-full or non-existent slot.	Booking is the core citizen-facing transaction that drives the entire vaccination rollout workflow.
FR-002	The system shall check real-time slot availability at the chosen vaccination center before confirming a booking (included by Book Vaccination Slot).	High	Pass: Booking is confirmed only if remaining capacity > 0 at the time of confirmation. Fail: Two citizens are confirmed for the same last-available slot.	Prevents over-booking and double-allocation of limited daily center capacity.
FR-003	The system shall enforce dose interval rules (e.g., minimum 28 days after Dose 1) before unlocking Dose 2 booking for a citizen (included by Book Vaccination Slot).	High	Pass: Dose 2 booking blocked if interval < 28 days. Fail: Early vaccination slot confirmed.	Clinically mandated spacing between doses must be enforced to ensure vaccine efficacy and citizen safety.
FR-004	The system shall allow a Vaccination Officer to generate a digitally signed, QR-coded vaccination certificate once a dose has been administered and recorded.	High	Pass: Certificate with valid QR is issued only after dose administration is recorded. Fail: Certificate generated for an unrecorded/unadministered dose.	A verifiable certificate is the end deliverable citizens need for travel, employment, and compliance proof.
FR-005	The system shall allow a Vaccination Officer to verify a citizen's vaccination certificate by scanning its QR code and validating the digital signature.	Medium	Pass: Valid, unaltered certificates are marked verified; tampered or expired ones are rejected. Fail: A forged QR code is accepted as valid.	Enables trustworthy on-site verification at checkpoints, workplaces, and travel gateways.

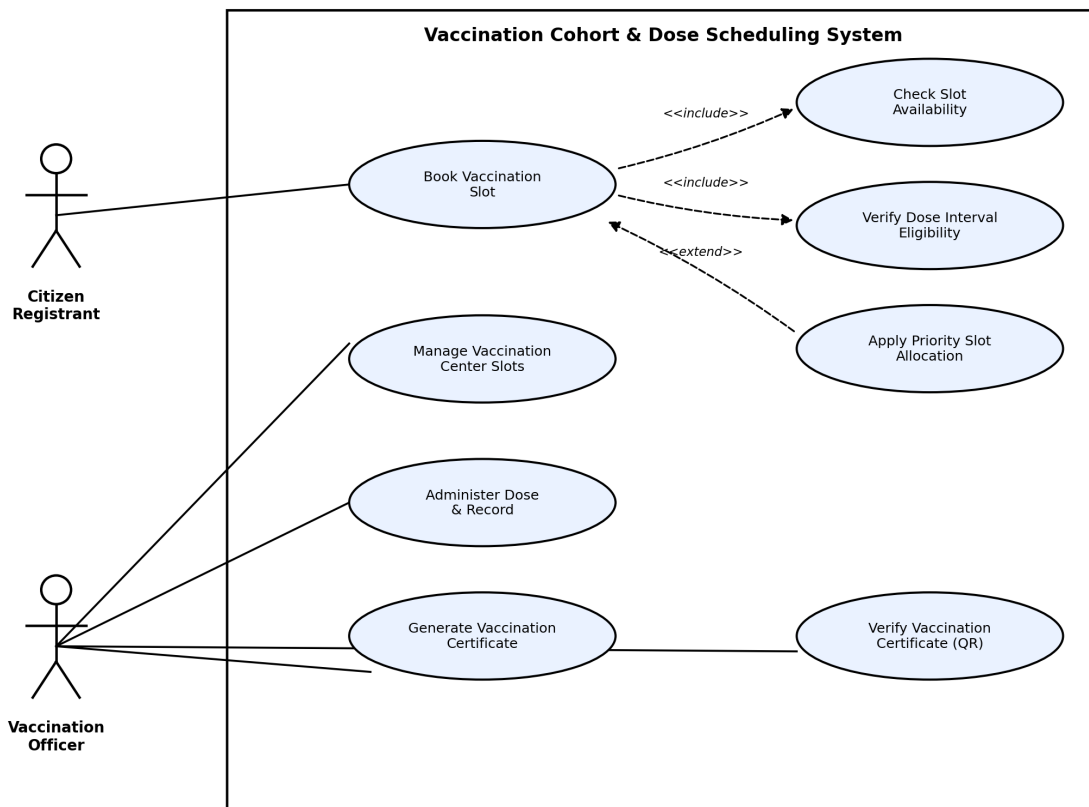
Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under 150 ms.	High	Pass: Benchmarking confirms sub-150ms verification under simulated peak load. Fail: Verification exceeds threshold or accepts an invalid signature.	Checkpoints (airports, offices) need near-instant verification to avoid queuing bottlenecks, and signature checks prevent forged certificates.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Reliability & Data Consistency	The system shall guarantee atomic, consistent slot booking such that no two concurrent requests can be confirmed for the same slot, and shall maintain 99.5% uptime during active vaccination drive hours (6 AM–10 PM).	Medium	Pass: Concurrency/load tests show zero double-bookings; uptime logs meet 99.5% over the drive period. Fail: A race condition produces a duplicate slot confirmation.	Large-scale rollouts see bursts of simultaneous booking attempts; consistency and availability directly affect public trust and health-camp logistics.

Deliverable 2: UML Use-Case Diagram

Actors: **Citizen Registrant** and **Vaccination Officer**. The diagram models the primary use cases inside the system boundary, with one «include» relationship (Book Vaccination Slot always includes Check Slot Availability and Verify Dose Interval Eligibility) and one «extend» relationship (Apply Priority Slot Allocation optionally extends Book Vaccination Slot for senior-citizen/comorbidity cohorts).



Deliverable 3: Use-Case Flow Specification

Use Case: Book Vaccination Slot

Actor(s)	Citizen Registrant (primary); System (Check Slot Availability, Verify Dose Interval Eligibility — included use cases)
Preconditions	1. The citizen is registered on the platform with a valid demographic/health ID. 2. If booking Dose 2 or later, the citizen's prior dose record (with administration date) already exists in the system. 3. At least one vaccination center in the citizen's eligible cohort is configured with active slots.
Postconditions	1. On success, a slot is reserved for the citizen at the chosen center, date, and time, and the center's remaining capacity is decremented by one. 2. A booking confirmation (with reference ID) is generated and made visible to the citizen. 3. On failure, no slot is reserved and system state remains unchanged.
Trigger	The Citizen Registrant selects "Book Vaccination Slot" for a specific dose number.

Main Success Scenario

1. The Citizen Registrant selects the dose number (e.g., Dose 2) and chooses a preferred vaccination center from the list of eligible centers.
2. The system executes «include» use case *Verify Dose Interval Eligibility*, confirming that at least the minimum required interval (e.g., 28 days) has elapsed since the citizen's previous dose.
3. The system executes «include» use case *Check Slot Availability*, querying the selected center's remaining capacity for the citizen's chosen date and time window.
4. The system displays the available time slots at the selected center to the citizen.
5. The citizen selects a specific date and time slot and submits the booking request.
6. The system atomically reserves the slot, decrements the center's available capacity by one, and generates a unique booking reference ID.
7. The system displays a booking confirmation to the citizen, including the center address, date, time, and reference ID.
8. The use case ends successfully.

Alternate Flow A1: Selected Slot Becomes Unavailable

- A1.1 (branches from Main Success Scenario Step 5): Between the citizen viewing available slots and submitting the booking, another citizen's concurrent request fills the last remaining slot at that date/time.
- A1.2: The system re-validates availability at submission time via *Check Slot Availability* and detects that the chosen slot capacity has reached zero.
- A1.3: The system rejects the booking request without reserving a slot or generating a reference ID, and notifies the citizen that the slot is no longer available.
- A1.4: The system re-displays the updated list of currently available slots at the same or nearby centers.
- A1.5: The citizen selects a new slot, and the flow resumes at Main Success Scenario Step 5.